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SEAT No. :

PA-1342

[Total No. of Pages : 3

[5925]-521

S.E. (Computer / IT Engineering)
ENGINEERING MATHEMATICS - III
(2015 Pattern) (Semester - II) (207003)

Time : 2 Hours]

[Max. Marks : 50

Instructions to the candidates :

- 1) Neat diagrams must be drawn wherever necessary.
- 2) Figures to the right indicate full marks.
- 3) Your answers will be valued as a whole.
- 4) Use of electronic pocket calculator is allowed.
- 5) Assume suitable data if necessary.

Q1) a) Solve any two

[8]

i) $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = e^{2x}$

ii) $\frac{d^2y}{dx^2} - 4y = \cos(2x)$

iii) Solve by Method of variation of parameters $\frac{d^2y}{dx^2} - y = x$

b) Find the Fourier transform of $f(x) = 1 \quad |x| < 1$
 $= 0 \quad |x| > 1$

Hence, find the value of $\int_0^\infty \frac{\sin x}{x} dx$.

[4]

OR

P.T.O.

Q2) a) Solve : $\frac{dx}{dt} = 5x + y; \frac{dy}{dt} = y - 4x$. [4]

b) Solve any one : [4]

i) find z-transform of $f(k) = k^2, k \geq 0$

ii) find the inverse z-transform of

$$\frac{z}{(z-2)(z-3)}, |z| > 3$$

c) Solve the following difference equation

$$f(k+2) - 5f(k+1) + 6f(k) = 36; f(0) = f(1) = 0. \quad [4]$$

Q3) a) Find first four moments about mean, if the first four moments about median are 1, 2, 4 & 6, also find β_1 & β_2 . [4]

b) Fit least squares straight line to the following data and use it to find y at $x = 8$ [4]

x	1	3	5	6	10
y	2	0	4	7	8

c) A manufacturer knows from experience that the resistance of resistors he produces is normally distributed with mean 100 ohms and standard deviation 2 ohms. What percentage of resistors will have resistance between 98 ohms to 102 ohms? (Area between 0 to 1 is 0.3413). [4]

OR

Q4) a) A can hit the target 1 out of 4 times

B can hit the target 2 out of 3 times

C can hit the target 2 out of 4 times

find the probability that at least two hit the target. [4]

b) In a sample of 1000 cases, the mean of certain test is 14 and standard deviation is 2.5 Assuming the distribution to be normal find how many students score between 12 and 15.

(Given : $Z = 0.8, A = 0.2881, Z = 0.4, A = 0.1554$). [4]

c) Obtain the regression lines for the following data [4]

x	6	2	10	4	8
y	9	11	5	8	7

Q5) a) Find the directional derivative of $\phi = xy^2 + yz^3$ at $(1, -1, 1)$ along the vector $\bar{i} + 2\bar{j} + 2\bar{k}$. [4]

b) Show that $\bar{F} = (6xy + z^3)\bar{i} + (3x^2 - z)\bar{j} + (3xz^2 - y)\bar{k}$ is irrotational. Find scalar ϕ such that $\bar{F} = \nabla\phi$. [4]

c) Find the work done in force field $\bar{F} = 3x^2y\bar{i} + (x^3 + 2yz)\bar{j} + y^2\bar{k}$ in moving object from $(1, -2, 1)$ to $(3, 1, 4)$. [5]

OR

Q6) a) Find the directional derivative of $\phi = e^{2x} \cos yz$ at $(0, 0, 0)$ in the direction of tangent to the curve $x = a \sin t; y = a \cos t; z = at, at, t = \frac{\pi}{4}$. [4]

b) Show that (any one): [4]

$$\text{i) } \nabla^2 f(r) = \frac{d^2 f}{dr^2} + \frac{2}{r} \frac{df}{dr} \quad \text{ii) } \nabla^2 \left(\nabla \cdot \frac{\bar{r}}{r^2} \right) = \frac{2}{r^4}$$

c) Evaluate $\int_c \bar{F} \cdot d\bar{r}$ for $\bar{F} = 3x^2\bar{i} + (2xz - y)\bar{j} + z\bar{k}$ along the curve $x = 2t^2, y = t, z = 4t^2 - t$ from $t = 0$ to $t = 1$. [5]

Q7) a) If $u = 3x^2 - 3y^2 + 2y$, find v such that $f(z) = u + iv$ is analytic. Determine $f(z)$ in terms of z . [4]

b) Find the bilinear transformation which maps the points $z = 1, i, 2i$ on the points $w = -2i, 0, 1$ respectively. [4]

c) Evaluate $\oint_c \frac{4z^2 + z}{z^2 - 1} dz$, where c is the contour $|z - 1| = \frac{1}{2}$. [5]

OR

Q8) a) Determine the analytic function $f(z) = u + iv$ if $u = xy^2 - 3xy$. [4]

b) Find the bilinear transformation, which maps the points $0, -1, i$ of the z -plane on to the points $2, \infty, \frac{1}{2}(5 + i)$ of the w -plane. [4]

c) Evaluate $\oint_c \frac{z^2 - z + 1}{z - 1} dz$, where c is the circle $|z| = \frac{1}{2}$. [5]

